

Experiment 8: Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering

Experiment No.	8
Experiment Title	Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering ¹
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1 Objective

To apply K-Means, DBSCAN and Hierarchical Agglomerative Clustering (HAC) on the Human Activity Recognition (HAR) smartphone dataset. The number of clusters for K-Means is chosen using the elbow and silhouette methods, the DBSCAN parameters are tuned, and the linkage criteria for hierarchical clustering are compared. All three algorithms are evaluated with internal and external clustering metrics.

2 Problem Statement

The input is a set of 10,299 sensor windows, each described by 561 features from a smart-phone accelerometer and gyroscope. The activity labels are not used while clustering. The output is a cluster label for every window. The aim is to check how well the clusters formed by each algorithm match the six real activities (walking, walking upstairs, walking downstairs, sitting, standing and laying) and to compare the three algorithms.

3 Dataset Description

Dataset Name	Human Activity Recognition Using Smart-phones
Dataset Source	UCI Machine Learning Repository [1]
Number of Samples	10,299 windows from 30 volunteers (7,352 train + 2,947 test)
Number of Features	561 (time and frequency domain features per 2.56 s window)
Number of Classes	6 (used only for evaluation)
Missing Values	0
Train-Test Split	Not used. The given train and test parts were merged because clustering is unsupervised

¹GitHub Repository: <https://github.com/Isha666/ML-Laboratory>

4 Exploratory Data Analysis

Dataset Overview

The combined data has 10,299 rows and 561 features, along with the activity label and subject id. There are no missing values and no duplicate rows. All feature values already lie between -1 and 1 , because the dataset authors normalised them. The 561 features come from 17 signal groups, and the largest groups are the frequency domain features of body acceleration, acceleration jerk and gyroscope (79 features each).

Statistical Summary

Table 1 gives the summary of eight representative features.

Table 1: Statistical summary of representative features

Feature	Mean	Std	Min	Median	Max
tBodyAcc-mean()-X	0.2743	0.0676	-1	0.2772	1
tBodyAcc-std()-X	-0.6078	0.4387	-1	-0.9430	1
tGravityAcc-mean()-X	0.6692	0.5155	-1	0.9218	1
tBodyGyro-mean()-X	-0.0310	0.1832	-1	-0.0278	1
tBodyAccMag-mean()	-0.5482	0.4671	-1	-0.8746	1
fBodyAcc-mean()-X	-0.6228	0.4208	-1	-0.9456	1
fBodyAccMag-mean()	-0.5860	0.4453	-1	-0.8755	1
angle(X,gravityMean)	-0.4965	0.5112	-1	-0.7156	1

Missing Value Analysis

The dataset has 0 missing values in all 563 columns, so no imputation was needed.

Class Distribution

Table 2: Number of windows per activity

Activity	Windows	Percentage (%)
WALKING	1,722	16.72
WALKING UPSTAIRS	1,544	14.99
WALKING DOWNSTAIRS	1,406	13.65
SITTING	1,777	17.25
STANDING	1,906	18.51
LAYING	1,944	18.88

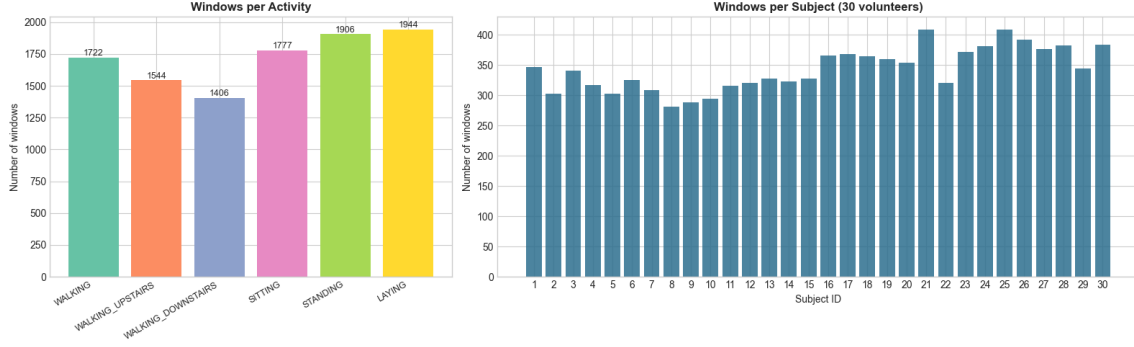


Figure 1: Number of windows per activity and per subject

Inference: The six activities are fairly balanced, from 1,406 windows for walking down-stairs to 1,944 windows for laying. No single class dominates, so the true labels form six groups of similar size. All 30 volunteers contribute windows. This balance makes it fair to compare the clusters with the true labels.

Feature Distribution

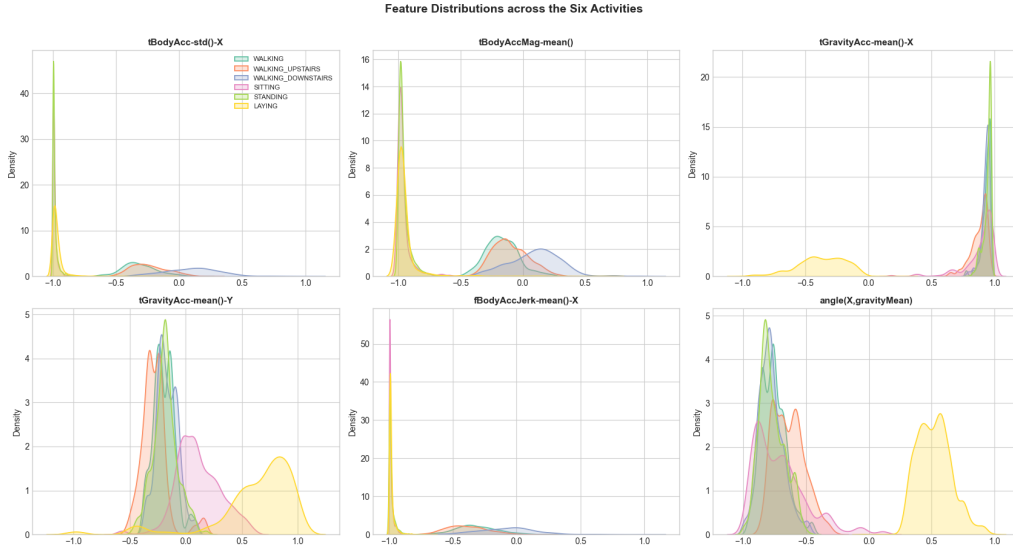


Figure 2: Distribution of six representative features for each activity

Inference: Body acceleration features such as `tBodyAcc-std()-X` and `tBodyAccMag-mean()` separate the moving activities from the static ones almost completely. Gravity related features such as `tGravityAcc-mean()-X` and `angle(X,gravityMean)` help to separate laying from sitting and standing. Inside each group the curves overlap a lot, especially for sitting and standing and for the three walking activities.

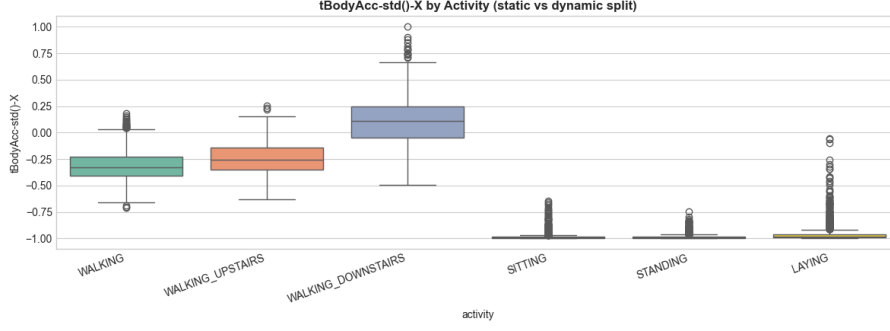


Figure 3: Box plot of $tBodyAcc-std()-X$ for each activity

Inference: The values for sitting, standing and laying stay close to -1 , while the three walking activities have clearly higher values. One feature alone is enough to split static and dynamic activities. The three walking activities overlap with each other, so this feature cannot tell them apart.

Correlation Matrix

Out of 157,080 feature pairs, 8,093 pairs (5.15%) have $|r| > 0.9$ and 22,815 pairs (14.52%) have $|r| > 0.8$. The mean absolute correlation over all pairs is 0.418.

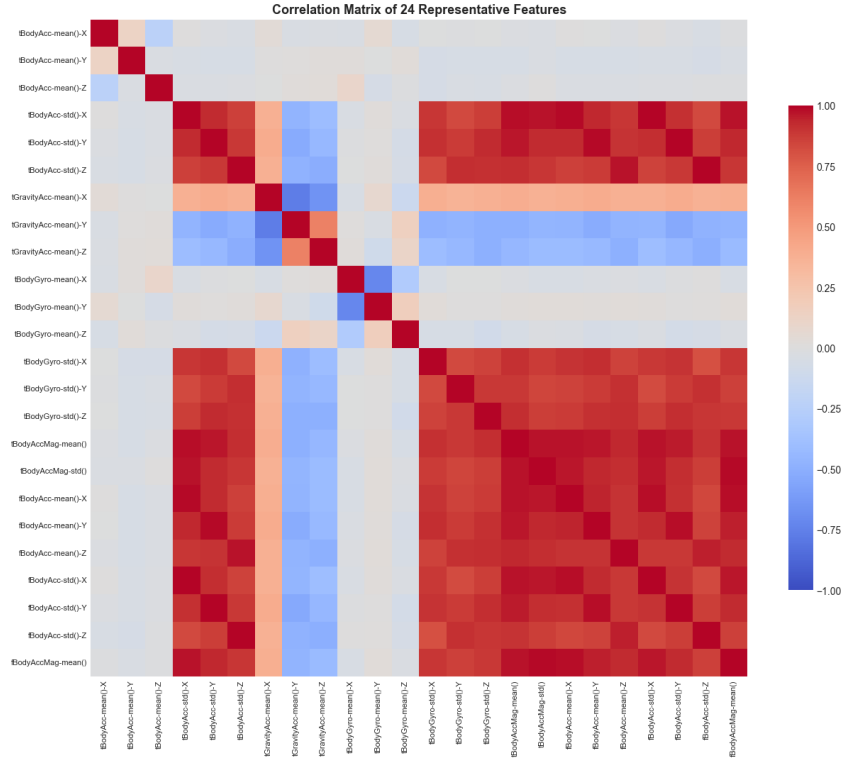


Figure 4: Correlation matrix of 24 representative features

Inference: The standard deviation and magnitude features of the body acceleration and gyroscope show strong positive correlation, seen as large dark red blocks. The mean features of body acceleration and gyroscope are almost uncorrelated with them. Many

features therefore carry repeated information. This supports reducing the dimensions with PCA before clustering.

5 Data Preprocessing

- **Missing value handling:** No missing values were found, so no imputation was done.
- **Encoding:** The activity labels are integers from 1 to 6 and were mapped to activity names. They are used only for evaluation and not for clustering.
- **Feature scaling:** `StandardScaler` was applied so that every feature has zero mean and unit variance. This stops a few wide range features from dominating the Euclidean distance.
- **Train-test split:** Not used. The 7,352 training windows and 2,947 test windows were merged into 10,299 windows.
- **Feature engineering:** The 561 features were already extracted by the dataset authors. A few repeated feature names were made unique. PCA was applied to keep 10 components (70.21% of the variance). This reduced space is used by all three algorithms. t-SNE [8] (perplexity 30) was applied on the PCA output only for 2D visualisation.

Table 3: Cumulative variance explained by PCA

Components	Cumulative Variance (%)
2	56.98
5	64.01
10	70.21
20	76.84
50	87.10
100	94.67
104	95.05

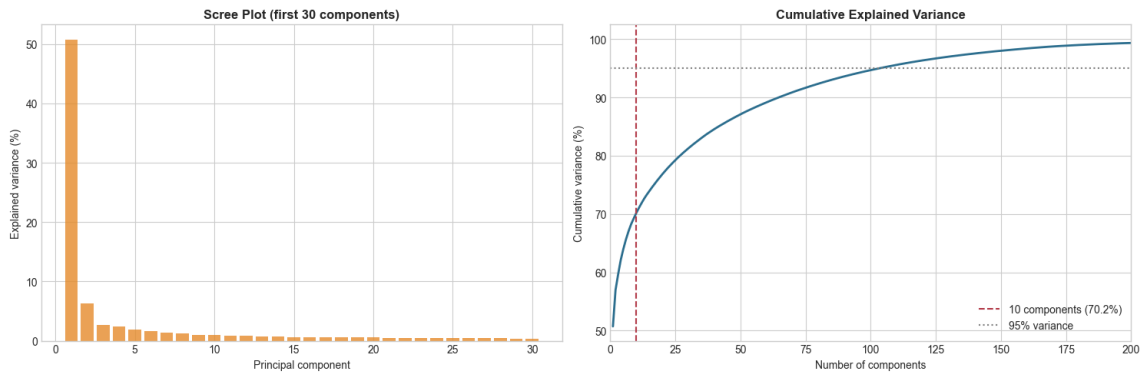


Figure 5: Scree plot and cumulative explained variance

Inference: The first 10 components explain 70.21% of the variance, and about 104 components are needed to reach 95%. The scree plot falls sharply after the first few components, so most of the information lies in a few directions. Ten components were kept because DBSCAN works poorly in hundreds of dimensions and because hierarchical clustering runs faster in a small space.

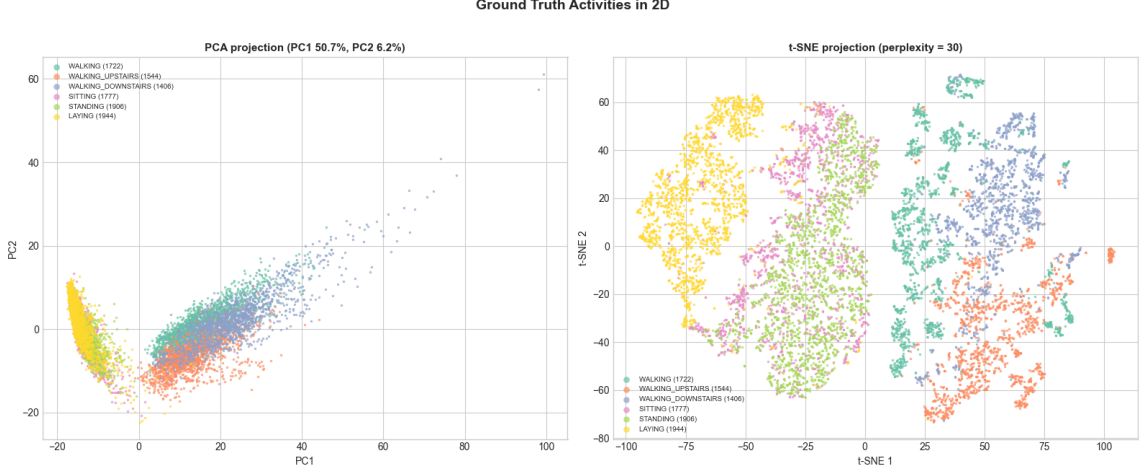


Figure 6: True activities in the PCA and t-SNE projections

Inference: In both projections the static activities (sitting, standing, laying) and the moving activities (three walking types) form two separate regions. Inside each region the classes touch or overlap, mainly sitting with standing and the three walking activities with each other. This shows what a good clustering should find, and it also shows that a perfect recovery of all six activities is unlikely.

6 Machine Learning Algorithm

Model A: K-Means Clustering

Working principle: K-Means [3] divides the data into k clusters. It starts with k centroids, assigns every point to the nearest centroid, moves each centroid to the mean of its points, and repeats until the assignments stop changing.

Mathematical formulation: The algorithm minimises the within-cluster sum of squares (WCSS, also called inertia):

$$J = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2$$

where C_i is the i -th cluster and μ_i is its centroid.

Advantages: Simple, fast and works well on large datasets.

Limitations: k must be chosen beforehand, the result depends on the starting centroids, it prefers round clusters of similar size, and outliers affect the centroids.

Model B: DBSCAN

Working principle: DBSCAN [4] groups points that are densely packed. A point is a core point if at least $minPts$ points (including itself) lie within a radius ϵ . Core points

that are close to each other join the same cluster, border points join the cluster of a nearby core point, and the remaining points are labelled as noise.

Mathematical formulation: The neighbourhood of a point p is

$$N_\epsilon(p) = \{ q : d(p, q) \leq \epsilon \}$$

and p is a core point if $|N_\epsilon(p)| \geq \text{minPts}$.

Advantages: No need to give the number of clusters, can find clusters of any shape, and detects noise.

Limitations: Sensitive to ϵ and minPts , does not handle clusters of very different density well, and performs poorly in high dimensions.

Model C: Hierarchical Agglomerative Clustering (Ward Linkage)

Working principle: Every point starts as its own cluster. At each step the two closest clusters are merged until one cluster remains. The merge history is shown as a dendrogram, and cutting the dendrogram at a chosen height gives the final clusters. With Ward linkage [5], the pair whose merge increases the within-cluster variance the least is merged.

Mathematical formulation: The Ward cost of merging clusters A and B is

$$\Delta(A, B) = \frac{|A||B|}{|A| + |B|} \|\mu_A - \mu_B\|^2$$

Advantages: The number of clusters can be decided after seeing the dendrogram, and Ward linkage gives compact clusters of balanced size.

Limitations: Time and memory grow quickly with the number of points, a merge cannot be undone, and the result changes a lot with the linkage criterion.

Evaluation Metrics

- **Silhouette score** [6]: $s = (b - a) / \max(a, b)$, where a is the mean distance to points in the same cluster and b is the mean distance to the nearest other cluster. Higher is better.
- **Davies-Bouldin index:** average similarity between each cluster and its most similar cluster. Lower is better.
- **Calinski-Harabasz index:** ratio of between-cluster to within-cluster spread. Higher is better.
- **Adjusted Rand Index (ARI)** [7] and **Normalized Mutual Information (NMI):** agreement between the clusters and the true activity labels. Higher is better.
- **Mapped accuracy:** clusters are matched to activities with the Hungarian algorithm and accuracy is computed. DBSCAN noise points are left out.

Internal metrics are computed in the 10-dimensional PCA space, and noise points are excluded for DBSCAN.

7 Experimental Setup

Python Version	3.13.5
Scikit-Learn Version [2]	1.x (see requirements.txt)
Other Libraries	NumPy, pandas, SciPy, Matplotlib, Seaborn
IDE	Jupyter Notebook
Operating System	macOS
Hardware	Apple Silicon Mac

8 Hyperparameter Selection

Parameter	Value
Scaling	StandardScaler
PCA components	10
t-SNE perplexity (visualisation only)	30
Random state	42
K-Means: values of k tried	2 to 8
K-Means: initialisations (<code>n_init</code>)	10
K-Means: selected k	6
DBSCAN: ϵ values tried	2.5 to 7.0 (step 0.5)
DBSCAN: $minPts$ values tried	5, 10, 20
DBSCAN: selected ϵ and $minPts$	6.0 and 20
Hierarchical: linkage	Ward (Euclidean distance)
Hierarchical: number of clusters	6

9 Results

K-Means: Elbow Method

Table 4 gives the WCSS and the silhouette score for k from 2 to 8, and Table 5 gives the remaining metrics for the same models.

Table 4: K-Means Elbow Method Results

Number of Clusters (k)	WCSS (Inertia)	Silhouette Score
2	1,552,171	0.5606
3	1,204,169	0.4734
4	1,067,331	0.2967
5	942,608	0.2727
6	866,550	0.2806
7	815,451	0.2733
8	761,871	0.2480

Table 5: Additional metrics for K-Means with different k

k	WCSS drop (%)	Davies-Bouldin	Calinski-Harabasz	ARI	NMI
2	-	0.6941	16,613.27	0.3296	0.5455
3	22.42	1.0840	12,193.97	0.3314	0.5188
4	11.36	1.3725	9,610.61	0.2988	0.4698
5	11.69	1.3996	8,501.41	0.2909	0.4603
6	8.07	1.2874	7,578.05	0.4271	0.5621
7	5.90	1.2442	6,817.59	0.4140	0.5564
8	6.57	1.3754	6,357.41	0.4101	0.5438

Selection of k : The WCSS drops sharply up to $k = 3$ (22.42% from $k = 2$ to $k = 3$) and the reduction falls to about 6% after $k = 6$. The silhouette score is highest at $k = 2$, which is the static versus moving split, but this is a coarse solution that merges three activities on each side. Among the finer solutions the silhouette has a small local peak at $k = 6$ (0.281), and $k = 6$ also gives the best ARI (0.427) and NMI (0.562). The elbow region, the local silhouette peak and the six real activities all point to $k = 6$, so $k = 6$ was selected. The $k = 2$ solution is kept for comparison.

DBSCAN: Parameter Tuning

The grid in Table 6 was run on the PCA space. The silhouette is computed on the non-noise points. The chosen setting has the highest silhouette among the settings with less than 10% noise and at least 3 clusters, so the true labels were not used for the choice. This gave $\epsilon = 6.0$ and $minPts = 20$, with 3 clusters, 8.06% noise, silhouette 0.5614, ARI 0.3210 and NMI 0.4996. The core, border and noise points were 8,545, 924 and 830. The cluster sizes were 5,520, 3,915 and 34, and a further setting ($\epsilon = 5.5$, $minPts = 10$) gave 7 clusters with 7.14% noise, but the extra clusters were only small fragments of the two large groups.

Table 6: DBSCAN tuning grid

ϵ	$minPts$	Clusters	Noise (%)	Silhouette	ARI	NMI
2.5	5	53	52.59	-0.1814	0.1857	0.3309
2.5	10	12	59.90	-0.0626	0.1519	0.2858
2.5	20	1	66.30	n/a	0.1110	0.2343
3.0	5	45	38.62	-0.0132	0.2399	0.3880
3.0	10	19	47.88	0.5139	0.2183	0.3725
3.0	20	4	55.39	0.5714	0.1946	0.3324
3.5	5	30	27.41	0.2101	0.2931	0.4409
3.5	10	10	34.95	0.4431	0.2655	0.4134
3.5	20	9	44.55	0.5207	0.2394	0.4017
4.0	5	30	18.08	0.1654	0.3082	0.4558
4.0	10	8	24.87	0.4311	0.3069	0.4556
4.0	20	5	33.31	0.4839	0.2760	0.4233
4.5	5	18	11.20	0.2691	0.3172	0.4822
4.5	10	9	16.81	0.3796	0.3139	0.4683
4.5	20	3	23.63	0.5159	0.3129	0.4644

ϵ	$minPts$	Clusters	Noise (%)	Silhouette	ARI	NMI
5.0	5	13	7.17	0.4039	0.3221	0.4992
5.0	10	6	10.81	0.4467	0.3172	0.4849
5.0	20	3	16.88	0.4735	0.3128	0.4704
5.5	5	10	4.67	0.4272	0.3252	0.5139
5.5	10	7	7.14	0.4834	0.3229	0.5035
5.5	20	4	11.61	0.4495	0.3141	0.4807
6.0	5	7	3.23	0.4176	0.3274	0.5245
6.0	10	4	5.01	0.5333	0.3249	0.5150
6.0	20	3	8.06	0.5614	0.3210	0.4996
6.5	5	2	2.34	0.3554	0.0025	0.0151
6.5	10	3	3.39	0.3039	0.0049	0.0316
6.5	20	3	5.88	0.5532	0.3241	0.5117
7.0	5	2	1.86	0.4766	0.0019	0.0135
7.0	10	2	2.62	0.3183	0.0032	0.0188
7.0	20	2	4.15	0.3708	0.0059	0.0355

Hierarchical Clustering: Linkage Comparison

Table 7 compares four linkage criteria with six clusters each.

Table 7: Agglomerative clustering with different linkage criteria (6 clusters)

Linkage	Largest cluster	Silhouette	Davies-Bouldin	Calinski-Harabasz	ARI	NMI
Ward	3,344	0.2626	1.3053	6,712.62	0.4897	0.6096
Complete	9,912	0.2076	1.0586	379.28	0.0036	0.0416
Average	5,573	0.5327	0.8401	4,433.34	0.3296	0.5425
Single	10,293	0.3701	0.3107	31.86	0.0001	0.0013

Comparison of the Three Algorithms

Table 8: Final metrics of the three clustering algorithms

Metric	K-Means ($k = 6$)	DBSCAN	Hierarchical (Ward)
Number of clusters	6	3	6
Noise (%)	0	8.06	0
Silhouette score	0.2806	0.5614	0.2626
Davies-Bouldin index	1.2874	0.6224	1.3053
Calinski-Harabasz index	7,578.05	10,399.27	6,712.62
Adjusted Rand Index	0.4271	0.3210	0.4897
Normalized Mutual Information	0.5621	0.4996	0.6096
Mapped accuracy	0.5459	0.3640	0.5960

Table 9: Per-activity recall (%) after mapping clusters to activities

Activity	K-Means	DBSCAN	Hierarchical (Ward)
WALKING	4.5	100.0	62.3
WALKING UPSTAIRS	80.1	2.4	81.5
WALKING DOWNSTAIRS	63.2	0.0	9.0
SITTING	22.0	0.0	13.7
STANDING	76.7	100.0	94.3
LAYING	80.7	0.0	84.4

10 Result Visualization

K-Means

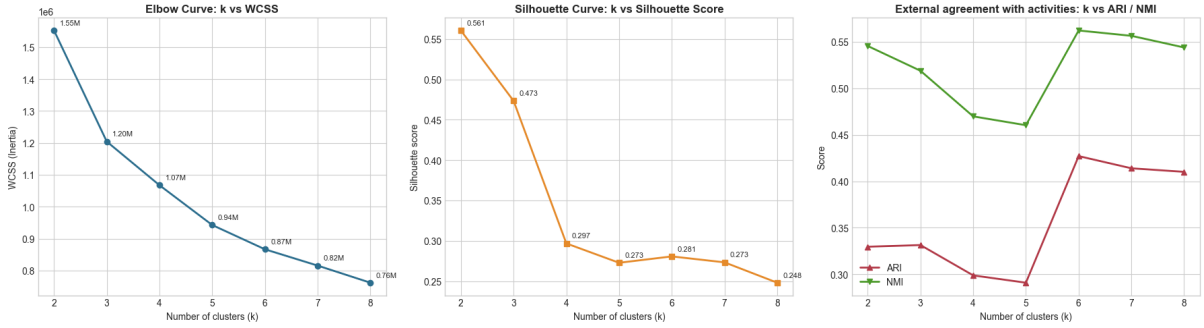


Figure 7: Elbow curve (k vs WCSS), silhouette curve (k vs silhouette score) and ARI/NMI for $k = 2$ to 8

Inference: The WCSS falls from 1.55 million at $k = 2$ to 0.76 million at $k = 8$, with the largest drop (22.4%) from $k = 2$ to $k = 3$ and drops of only about 6% after $k = 6$. The silhouette is highest at $k = 2$ (0.561) and has a small local peak at $k = 6$ (0.281). ARI and NMI both peak at $k = 6$, so the three curves support $k = 6$ as the final choice.

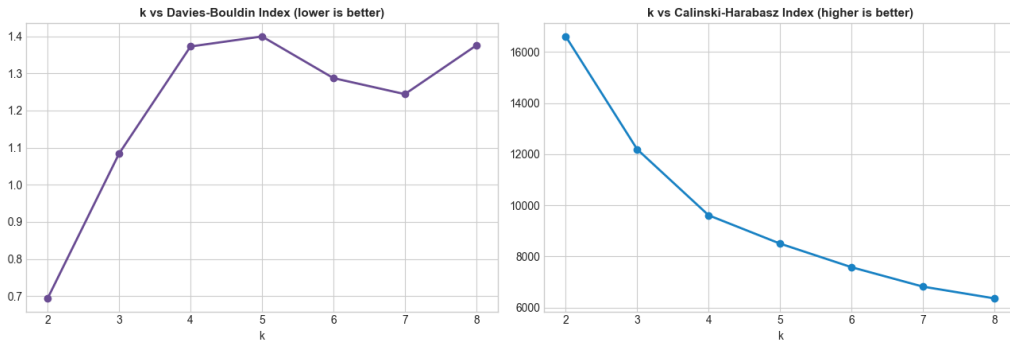


Figure 8: Davies-Bouldin index and Calinski-Harabasz index for different k

Inference: Both indices prefer the coarse $k = 2$ solution, where the Davies-Bouldin value is lowest (0.694) and the Calinski-Harabasz value is highest (16,613). The Davies-Bouldin index improves a little at $k = 6$ (1.287) compared with $k = 5$ (1.400). The

Calinski-Harabasz index keeps decreasing as k grows, so it cannot be used alone to choose k .

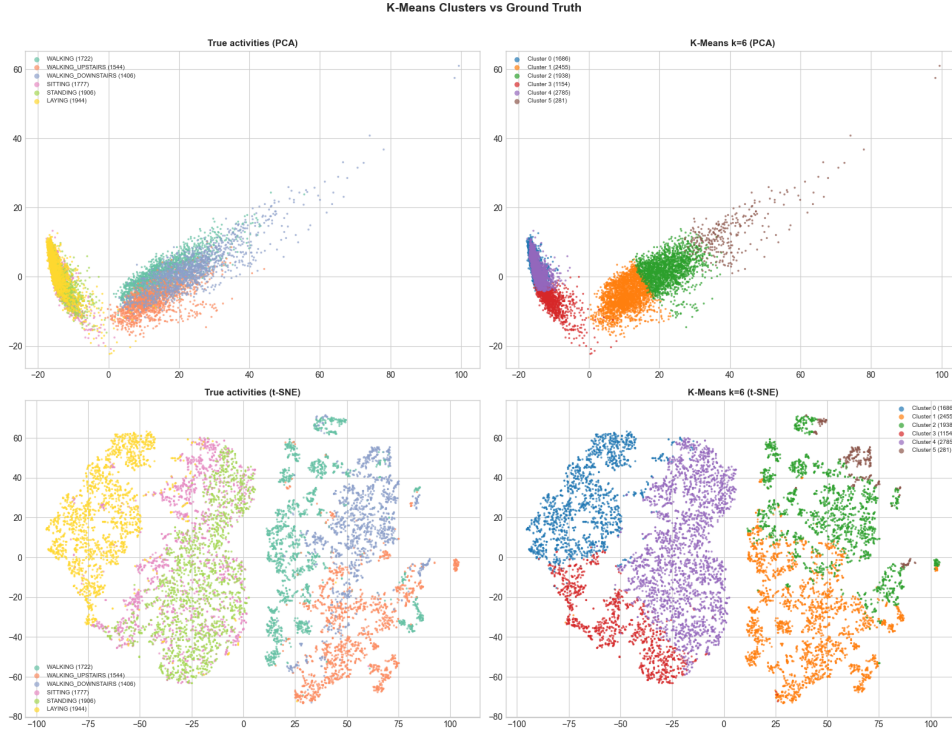


Figure 9: K-Means clusters ($k = 6$) compared with the true activities in PCA and t-SNE views

Inference: K-Means forms six clusters with sizes from 281 to 2,785 windows. One cluster holds most of the laying windows (1,568 of 1,944), while the walking windows are spread over three clusters with a lot of mixing. Sitting and standing are shared between two clusters and overlap heavily.

DBSCAN

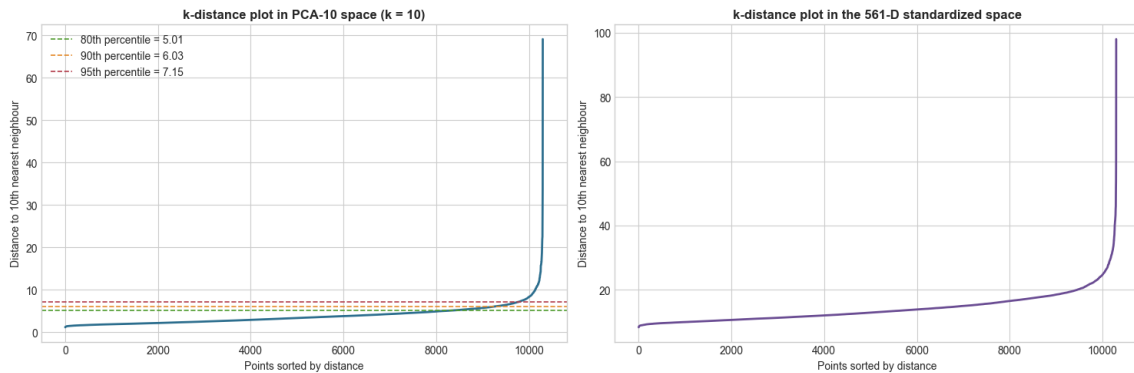


Figure 10: k-distance plot ($k = 10$) in the PCA space and in the full 561-dimensional space

Inference: In the PCA space the distance to the 10th neighbour rises slowly and then sharply. The 80th to 95th percentile values are 5.01 to 7.15, so the ϵ grid was chosen

between 2.5 and 7.0. In the full space the distances are much larger (median 13.07) and vary less (max to min ratio 11.7 against 61.8 in PCA space), so there is little density contrast. DBSCAN in the full space gave either one cluster or 38.5% noise.

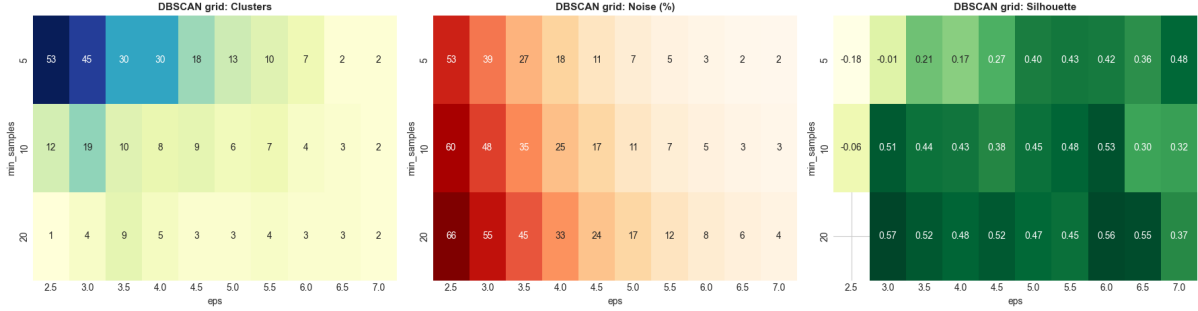


Figure 11: DBSCAN grid: number of clusters, noise percentage and silhouette for each ϵ and $minPts$

Inference: Small ϵ values label a large share of the data as noise (52.59% at $\epsilon = 2.5$, $minPts = 5$) and give many small clusters. Large ϵ values with small $minPts$ merge the data into 2 or 3 clusters. The setting $\epsilon = 6.0$, $minPts = 20$ gives the best balance with 3 clusters, 8.06% noise and the highest silhouette among the accepted settings.

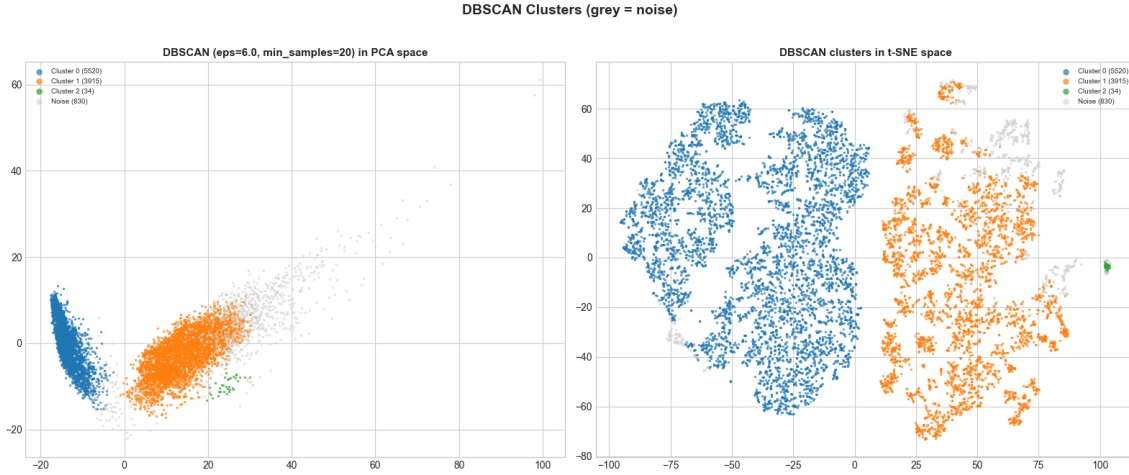


Figure 12: DBSCAN clusters ($\epsilon = 6.0$, $minPts = 20$) in PCA and t-SNE views, noise in grey

Inference: DBSCAN finds two large dense groups (5,520 and 3,915 windows) and one small cluster of 34 windows. The two large groups correspond to the static and the moving activities. It marks 830 windows (8.06%) as noise, and 723 of them come from the walking activities, mostly walking downstairs (414 windows).

Hierarchical Clustering

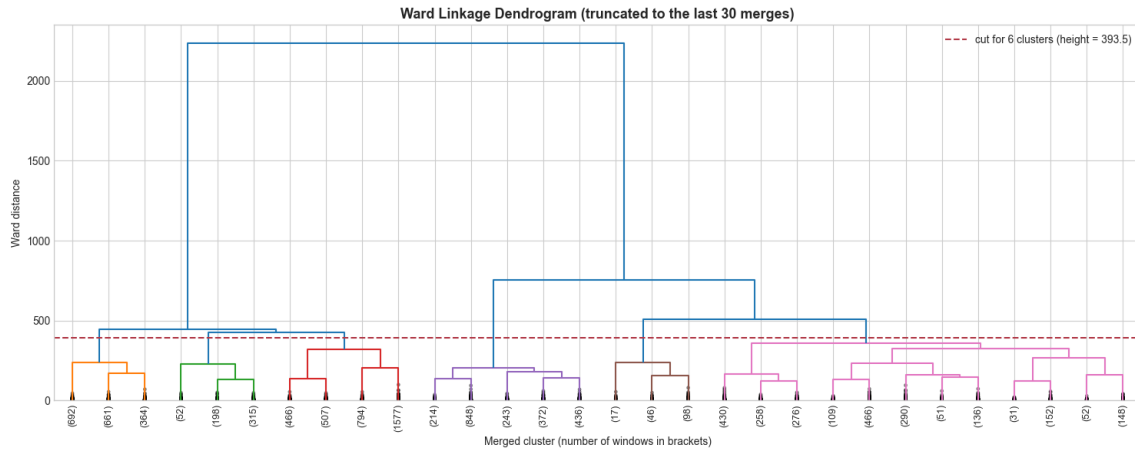


Figure 13: Ward linkage dendrogram truncated to the last 30 merges

Inference: The last merge happens at a height of about 2,236, far above the next merge at 754. This shows that the split into static and moving activities is the strongest structure in the data. Cutting at a height of 393.5 gives six clusters of sizes 3,344, 2,399, 2,113, 1,717, 565 and 161.



Figure 14: Effect of the linkage criterion in the t-SNE view

Inference: Ward linkage gives clusters of balanced size. Complete, average and single linkage each give one or two very large clusters with a few tiny ones. Single linkage puts 10,293 windows in one cluster and leaves five clusters with 1 or 2 windows each, which is the chaining effect.

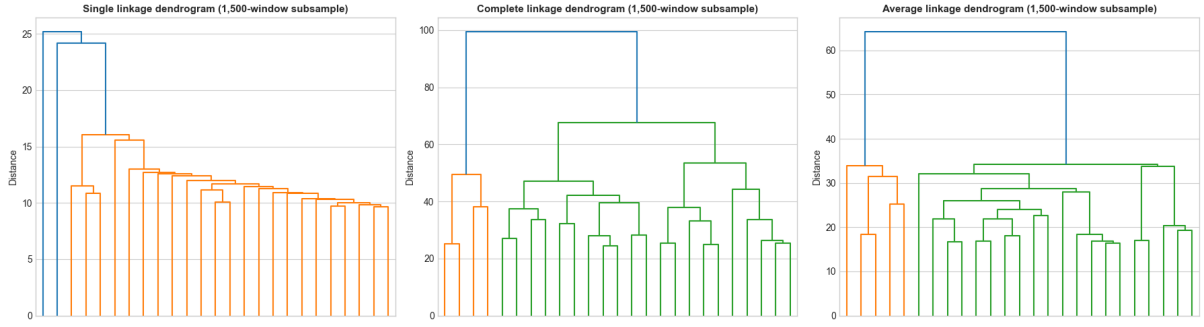


Figure 15: Dendrograms of single, complete and average linkage on a 1,500-window subsample

Inference: The single linkage dendrogram has a staircase shape, where points join one large group one after another and a few outliers join last. Complete and average linkage have a very large final merge that separates a small group from the rest. None of them shows the balanced splits seen with Ward linkage.

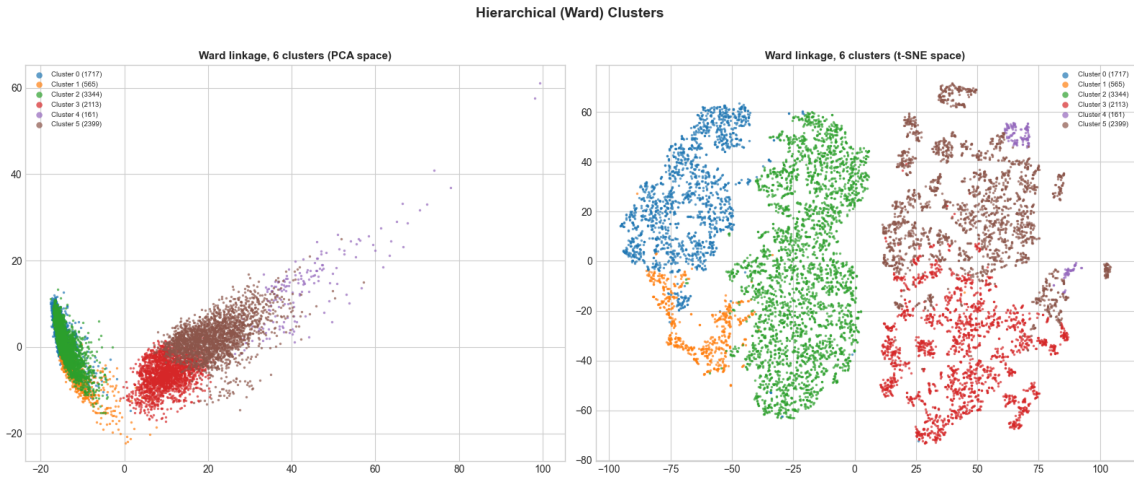


Figure 16: Ward clusters (6 clusters) in PCA and t-SNE views

Inference: Ward linkage puts most of the laying windows (1,641 of 1,944) in one cluster and keeps sitting and standing together in one large cluster of 3,344 windows. The walking windows split into a cluster dominated by walking upstairs and a cluster that mixes walking and walking downstairs. This is close to the structure of the true activities.

Comparison of Algorithms

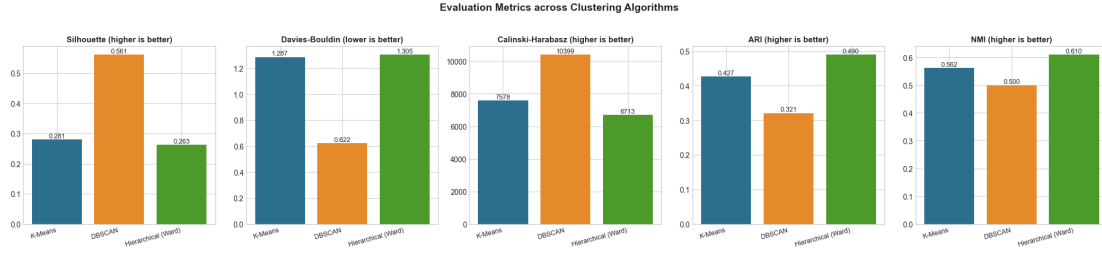


Figure 17: Bar plots of the five evaluation metrics for the three algorithms

Inference: DBSCAN has the best silhouette (0.561), Davies-Bouldin index (0.622) and Calinski-Harabasz index (10,399) among the three final models. Ward linkage has the best ARI (0.490) and NMI (0.610), followed by K-Means. The internal and external metrics disagree because the internal metrics reward the two large groups, while the external metrics compare against six activities.

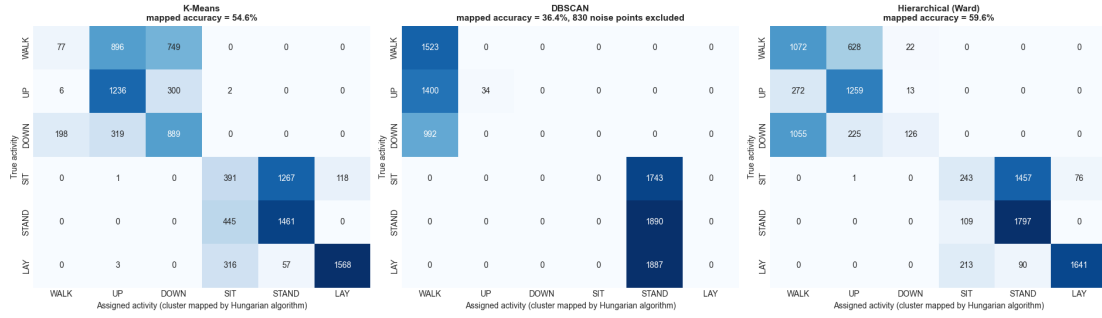


Figure 18: Confusion matrices after mapping clusters to activities

Inference: The mapped accuracy is 59.6% for Ward, 54.6% for K-Means and 36.4% for DBSCAN (noise points excluded). Most errors in all three are between sitting and standing and among the three walking activities. DBSCAN has only three clusters, so several activities cannot get a cluster of their own.

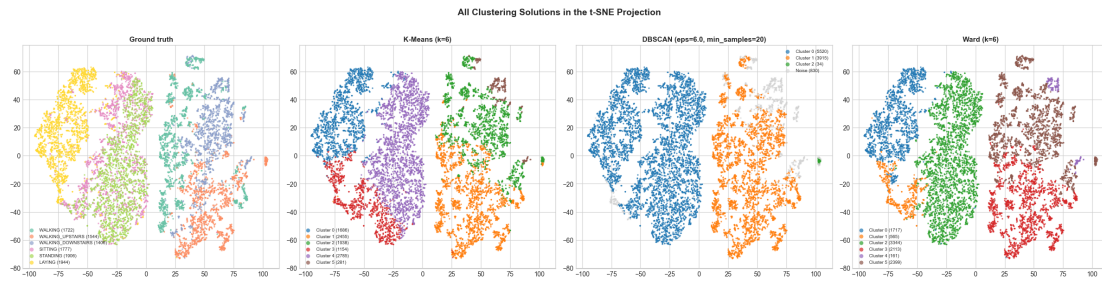


Figure 19: Ground truth and all three clustering results in the t-SNE view

Inference: K-Means and Ward divide the t-SNE map into regions of similar size to the true activities. DBSCAN shows two large blobs with grey noise points at the edges. None of the solutions reproduce the boundaries between the touching activities.

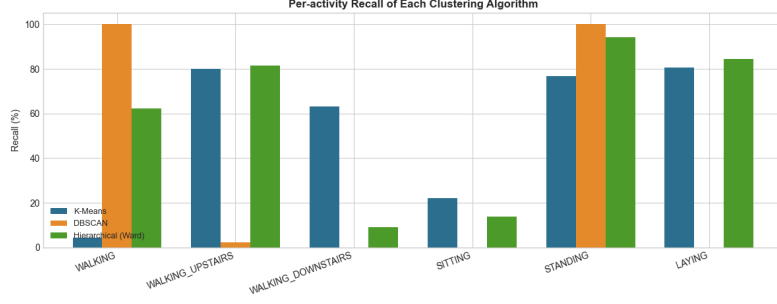


Figure 20: Per-activity recall of each algorithm

Inference: Ward recovers standing (94.3%), laying (84.4%) and walking upstairs (81.5%) best, while K-Means does better on walking downstairs (63.2% against 9.0%). The DBSCAN recall is either 100% or close to 0, because its two large clusters map to only walking and standing. Sitting has low recall in all three algorithms.

11 Discussion

Which algorithm produced the most meaningful clusters? Hierarchical clustering with Ward linkage matched the six activities best (ARI 0.490, NMI 0.610, mapped accuracy 59.6%), followed by K-Means with $k = 6$ (ARI 0.427, NMI 0.562). DBSCAN has the best internal scores, but only because it found the two-group split plus one tiny cluster. K-Means with $k = 2$ has a similar silhouette (0.561), so a high silhouette here does not mean the activities were recovered. No algorithm separated sitting from standing or the three walking activities, because these overlap in the feature space.

How sensitive was K-Means to the choice of k ? It was very sensitive. The silhouette dropped from 0.561 at $k = 2$ to 0.297 at $k = 4$ and stayed between 0.25 and 0.28 for $k = 5$ to 8. The ARI changed from 0.33 at $k = 2$ to 0.29 at $k = 4$ and 5, then rose to 0.43 at $k = 6$. K-Means also depended on the starting centroids. With a single random start, 6 of 10 seeds ended in a worse solution (WCSS from 877,913 to 889,786 and ARI from 0.285 to 0.349) than the best one (WCSS 866,550, ARI 0.427), so `n_init` was set to 10. Running K-Means in the 10 component PCA space gave almost the same result as the full space (ARI 0.427 against 0.420) and was faster (0.10 s against 0.73 s).

Did DBSCAN detect noise or small clusters effectively? It detected noise reasonably well. Of the 830 noise windows, 723 came from walking activities, and 29% of the walking downstairs windows were marked as noise. These windows are more variable and lie in sparse regions. Small cluster detection was limited: only one small cluster (34 walking upstairs windows) was found, and lowering ϵ mostly turned points into noise. DBSCAN also worked only after PCA. In the full 561-dimensional space it either merged everything into one cluster or marked 38.5% of the windows as noise.

How does the linkage choice affect hierarchical clustering? It had a very large effect. Single linkage chained almost all windows into one cluster (ARI 0.000), complete linkage gave one cluster of 9,912 windows (ARI 0.004), and average linkage found only the static and moving split (ARI 0.330). Ward linkage was the only one that gave balanced clusters and it had the best ARI (0.490) and NMI (0.610), because it merges the pair that adds the least variance.

Which internal metric best matched the visual intuition? The Calinski-Harabasz index matched best. It gave very low values to the single (31.9) and complete

(379) linkage results, which look wrong in the t-SNE plots. Silhouette and Davies-Bouldin were misleading for these results: single linkage got a silhouette of 0.370 and the best Davies-Bouldin value (0.311) although it is one giant cluster with a few single points. All internal metrics preferred the two-group view, while ARI and NMI preferred six clusters, which shows that the geometry of this dataset has two strong groups and not six.

Strengths and limitations: All three algorithms were run on the same standardized PCA space and compared with five metrics, which makes the comparison fair. The main limitation is that PCA keeps only 70.21% of the variance and that clustering only uses distance, so activities with similar sensor patterns cannot be separated. Possible improvements are trying Gaussian mixture models or spectral clustering, using more principal components, or using per-subject normalisation.

12 Conclusion

K-Means, DBSCAN and Hierarchical Agglomerative Clustering were applied on the HAR dataset after standardization and PCA. K-Means with $k = 6$ was selected using the elbow curve, the silhouette curve and the number of activities. DBSCAN with $\epsilon = 6.0$ and $minPts = 20$ found two large groups, one small cluster and 8.06% noise. Ward linkage gave the best agreement with the true activities (ARI 0.490, NMI 0.610), K-Means was close behind and faster, and DBSCAN was useful for finding unusual windows but not for recovering the six activities. All algorithms separated static from moving activities well, but none could separate sitting from standing or the walking types cleanly.

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